

CJR Chatbot Plugin - Critical Fixes Applied

Date: December 16, 2025

Summary

Fixed critical fatal error that was breaking the website and added comprehensive error handling throughout the plugin.

Critical Issues Fixed

1. Fatal Error: Undefined constant “cjrChatbot” (Line 23 in chat-widget.php)

Problem:

```
<?php echo esc_html(cjrChatbot.strings.welcome); ?>
```

This code was trying to access a JavaScript variable (`cjrChatbot`) in PHP, which caused a fatal error.

Solution:

```
<?php esc_html_e('👋 Hello! How can I help you with the Center for Justice Research today?', 'cjr-chatbot'); ?>
```

Changed to use proper PHP translation function instead of trying to access JavaScript variable.

File: `includes/templates/chat-widget.php` (line 23)

2. Missing JavaScript Localization String

Problem:

The JavaScript code referenced `cjrChatbot.strings.rateLimit` (line 95 in `chatbot.js`) but this string was not defined in the `wp_localize_script` call.

Solution:

Added the missing string to the localization array:

```
'rateLimit' => __('⚠️ Too many requests. Please wait a moment and try again.', 'cjr-chatbot'),
```

File: `includes/class-frontend.php` (line 49)

Error Handling Improvements

3. Frontend Template Rendering

Added: Try-catch blocks and file existence checks to prevent fatal errors

File: `includes/class-frontend.php`

render_chat_widget() method:

```
try {
    $template_path = CJR_CHATBOT_PLUGIN_DIR . 'includes/templates/chat-widget.php';
    if (file_exists($template_path)) {
        include $template_path;
    } else {
        error_log('CJR Chatbot: Widget template not found at ' . $template_path);
    }
} catch (Exception $e) {
    error_log('CJR Chatbot: Error rendering widget - ' . $e->getMessage());
}
```

chatbot_shortcode() method:

```
try {
    $template_path = CJR_CHATBOT_PLUGIN_DIR . 'includes/templates/chat-widget.php';
    if (file_exists($template_path)) {
        include $template_path;
    } else {
        error_log('CJR Chatbot: Widget template not found at ' . $template_path);
        echo '<p>' . esc_html__('Chatbot widget unavailable.', 'cjr-chatbot') . '</p>';
    }
} catch (Exception $e) {
    error_log('CJR Chatbot: Error rendering shortcode - ' . $e->getMessage());
    echo '<p>' . esc_html__('Chatbot widget error.', 'cjr-chatbot') . '</p>';
}
```

4. Plugin Initialization Error Handling

Added: Try-catch block in dependency loading with admin notice on error

File: `includes/class-plugin.php`

```

private function load_dependencies() {
    try {
        // Core classes are autoloaded
        $this->admin = new CJR_Chatbot_Admin();
        $this->frontend = new CJR_Chatbot_Frontend();
        $this->rest_api = new CJR_Chatbot_REST_API();
        $this->indexer = new CJR_Chatbot_Indexer();
        $this->security = new CJR_Chatbot_Security();
    } catch (Exception $e) {
        error_log('CJR Chatbot: Failed to load dependencies - ' . $e->getMessage());
        add_action('admin_notices', function() use ($e) {
            echo '<div class="notice notice-error"><p>';
            echo esc_html__('CJR Chatbot plugin encountered an error:', 'cjr-chat-
bot') . ' ' . $e->getMessage();
            echo '</p></div>';
        });
    }
}

```

5. Batch Indexing Error Handling

Added: API key validation and per-post error handling

File: `includes/class-indexer.php`

Changes:

- Added API key validation at the start of batch indexing
- Wrapped individual post indexing in try-catch blocks
- Added detailed error logging for failed indexing operations
- Prevents one failed post from stopping the entire indexing process

```

public function ajax_batch_index() {
    try {
        // Check API key
        $api_key = get_option('cjr_chatbot_api_key', '');
        if (empty($api_key)) {
            wp_send_json_error(array('message' => __('API key not configured. Please
set it in Settings.', 'cjr-chatbot')));
        }

        // ... batch processing ...

        foreach ($posts as $post) {
            try {
                if ($this->index_single_post($post->ID, $post)) {
                    $success++;
                } else {
                    $failed++;
                }
            } catch (Exception $e) {
                error_log('CJR Chatbot: Error indexing post ' . $post->ID . ' - ' .
$e->getMessage());
                $failed++;
            }
            // ...
        }
    } catch (Exception $e) {
        error_log('CJR Chatbot: Batch index error - ' . $e->getMessage());
        wp_send_json_error(array('message' => __('Indexing error: ', 'cjr-chatbot') .
$e->getMessage()));
    }
}

```

Benefits of These Fixes

1. **No More Fatal Errors:** The plugin will never break the website, even if errors occur
 2. **Better Debugging:** All errors are logged to PHP error log for easier troubleshooting
 3. **Graceful Degradation:** If something goes wrong, users see helpful messages instead of fatal errors
 4. **Improved Indexing:** Indexing now validates API key upfront and continues even if individual posts fail
 5. **User-Friendly:** Clear error messages guide users to fix configuration issues
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Testing Recommendations

1. **Test Widget Display:**
 - Clear browser cache
 - Visit the frontend of your site
 - Verify the chat widget appears and can be toggled
 - Check browser console for any JavaScript errors
2. **Test Chat Functionality:**
 - Click the chat button to open the widget

- Send a test message
- Verify you get a response (requires API key to be configured)

3. **Test Indexing:**

- Go to WP Admin → CJR Chatbot → Settings
- Configure your RouteLLM API key if not already set
- Click “Start Indexing” button
- Verify the progress bar works and shows success/failure counts
- Check for any error messages

4. **Test Q&A Management:**

- Go to WP Admin → CJR Chatbot → Q&A Management
- Add a test Q&A entry
- Verify it saves successfully
- Test editing and deleting Q&A entries

5. **Test Error Handling:**

- Temporarily remove the API key from settings
- Try to use the chatbot
- Verify you see a friendly error message instead of a fatal error
- Try indexing without API key
- Verify you see a proper error message

Files Modified

1. `includes/templates/chat-widget.php` - Fixed fatal error on line 23
2. `includes/class-frontend.php` - Added error handling and missing localization string
3. `includes/class-plugin.php` - Added error handling in dependency loading
4. `includes/class-indexer.php` - Added comprehensive error handling in batch indexing

Next Steps

1. **Deploy to Staging:** Upload the fixed plugin to your staging site
2. **Test Thoroughly:** Follow the testing recommendations above
3. **Configure API Key:** Ensure your RouteLLM API key is properly set
4. **Run Indexing:** Index your content to enable semantic search
5. **Monitor Logs:** Check PHP error logs for any remaining issues
6. **Deploy to Production:** Once stable on staging, deploy to production

Support

If you encounter any issues after applying these fixes:

1. Check the PHP error log for detailed error messages
2. Verify the API key is correctly configured

3. Ensure all plugin files are uploaded correctly
 4. Clear WordPress and browser caches
 5. Deactivate and reactivate the plugin if needed
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Version History

- **v1.0.1** (2025-12-16): Critical bug fixes and error handling improvements
- **v1.0.0** (Initial): Original refactored version